Jiayi Zhao

+1 7344895116 zjiay@uw.edu LinkedIn Github

Education

University of Washington, Seattle **Sep. 2025 -**
Ph. D of Electrical and Computer Engineering
GPA: 3.79

University of Michigan, Ann Arbor **Sep. 2023 - May 2025**
Bachelor of Science and Engineering in Data Science & Engineering (Mathematics Minor)
GPA: 3.72/4.0

University of Michigan Shanghai Jiaotong University **Sep. 2021 - August 2025**
Bachelor of Engineering in Electrical and Computer Engineering
GPA: 3.28/4.0

Course Modules: Robot Control, Control Theory for Biological Sensorimotor Systems, Numerical Methods, Natural Language Processing, Computer Vision, Entropy and Information, Data Structures and Algorithms, Introduction to Computer Organization, Applications of Stochastic Processes

Interests

- Machine learning
- Data analysis
- Multi-agent systems
- Distributed Controls
- Convex Optimization

Publications

Human Balancing on a Log: A Switched Multi-Layer Controller
Jiayi Zhao, Mo Yang, Jing Shuang (Lisa) Li
Area: Bio-mechanical control, nonlinear control systems, switched and hybrid systems.
Accepted & Presented ACC 2025.

Projects

Optimistic Gradient Descent-Ascent for Multi-Agent Skew-Symmetric Games **May 2026 - Jun. 2026**

- Introduced matrix-valued preconditioning beyond scalar OGDA to improve stability in multi-agent game dynamics
- Studied the convergence and stability behavior of Optimistic Gradient Descent-Ascent in multi-agent zero-sum games with skew-symmetric interaction matrices
- Derived spectral stability conditions using companion-matrix analysis, perturbation expansion, and eigenvalue properties
- Implemented Python simulations for 10-agent games under constant, decaying, and noisy step-size settings

Matrix-Valued Equivalence between Optimism and Augmented Lagrangian Methods **Mar. 2026 - Jun. 2026**

- Developed a matrix-valued correction framework connecting optimistic gradient updates with augmented Lagrangian dynamics
- Analyzed how structured matrix corrections can improve stability when constraint Jacobians have heterogeneous scales or ill-conditioned directions
- Derived linearized state-space representations to compare augmented, optimistic, and hybrid update channels
- Implemented Python simulations and achieved higher outer loop convergence rate than tested external solvers.

Switched multi-Layer controller for human postural balancing on uneven terrain **Feb. 2024 - Oct. 2024**

- Designed a three-case multi-layer controller with the inspiration from the human motor-sensory feedback control loop
- Developed a fully physically realistic model; designed and simulated the model using MATLAB
- Verified the Controller in MATLAB and tested noise-resistance

Large Able-bodied Dataset Standardization **May 2024 - Oct. 2024**

- Designed python pipeline for processing multiple datasets; integrated multiple non-parquet formatted datasets into a complete compatible parquet file

- Assisted in setting the standard of the data
- Verified the data with OpenSim simulation

Micro Spiking Neural Network Controller **Sep. 2023 - May 2024**

- Designed and implemented a RL-based controller for single leg under SNN architecture
- Designed and trained the model with python and torch
- Simulated the model in python

Passive Exoskeleton CTM Tennis Elbow Rehabilitation Device **Sep. 2023 - May 2024**

- Designed and optimized the Constrained Torque Model (CTM) based on detailed force and physics analysis of the Polycarbonate (PC) material in elastic deformation
- Designed the overall device designed with Solidworks
- Manufactured the model with 3D printing and performed effectiveness and comfort testing on participants

Experience

Chairman Consultant **Sep. 2022 - Aug. 2023**

SJTU Student Science, Technology and Innovation Association

- Planned and launched the freshman's AI contest
- Organized and coordinated the freshman's Robotic competition

Honors and Awards

University Honors **2024**

University of Michigan Dean's list. Recognizes exceptional academic achievement across the entire university.

Dean's List **2024**

University of Michigan Dean's list. Recognizes high academic performance within specific college or school.

University Honors **2023**

University of Michigan Dean's list. Recognizes exceptional academic achievement across the entire university.

Dean's List **2023**

University of Michigan Dean's list. Recognizes high academic performance within specific college or school.

Honorable Mention in MCM **2023**

A mathematical modelling contest for college students around the globe. Honor for top 21% participants

Finalist in IMMC (international) **2021**

A mathematical modelling contest for high school students around the globe. Honor for top 1.5% participants.

Meritorious Winner in HiMCM **2019**

A mathematical modelling contest for high school students around the globe. Honor for top 8% participants

Technical Skills

Languages: Python, Matlab, C++, C#

Developer Tools: VS Code, Git, OpenSim, Unity, SQL, LaTeX